Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
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FIRST SEMESTER 2022 FINAL ONLINE EXAMINATION

BACHELOR DEGREE – Year 2

INTRODUCTION TO PROGRAMMING IN JAVA

TIME ALLOWED: 1.5 Hours

INSTRUCTIONS TO CANDIDATES

1、 **Total marks available are 100. This will count for 40% of the final module mark.**

2、 **Answer all questions.**

3、 **All questions should be answered on the Learning Mall Quiz Page.**

4、 **All the answers must be in English.**

Part 1 Multiple-Choice Questions: 50 marks, 25 Questions

1. In Java, which line correctly declares the class that contains the program entry point?
   A. public static void Hello {
   B. class main(String[] args) {
   C. public class Hello {
   D. public void class Hello()

2. What is printed by the code below?

```
int a = 7, b = 3;
System.out.println(a / b);
```

   A. 2.3333333
   B. 2
   C. 3
   D. Compile error

3. Which expression is TRUE exactly when **a** and **b** differ (XOR), assuming boolean a, b?
   A. (a && b) || (!a && !b)
   B. (!a && b) || (a && !b)
   C. (a || b) && (a && b)
   D. !(a || b)

4. Consider:

```
int i = 1, val = 1;
while (i <= n) {
 System.out.println(val);
 val = val * 2;
 i = i + 1;
}
```

   Which bug appears if the loop condition is mistakenly written i < n?
   A. Compile error
   B. Infinite loop
   C. Off-by-one (prints one fewer line)
   D. Integer overflow

5. Which for-loop runs **exactly** 10 times?
   A. for (int i=10; i<20; i++)

B. for (int j=30; j>20; j--)
C. for (int k=1; k<=10; k++)
D. for (int l=5; l<25; l+=2)

6. Which statement about Java **arrays** is correct?
   A. Their size can grow automatically.
   B. Access past length-1 throws an exception.
   C. Elements are uninitialized garbage values.
   D. length() is the method to get the size.
7. What is the **scope** of the loop counter i in for (int i=0; i<10; i++) { ... }?
   A. Everywhere in the class
   B. Only inside the loop body
   C. From its declaration to program end
   D. Inside main only
8. Which call converts the string "123" to an int?
   A. Integer.toString("123")
   B. Double.parseDouble("123")
   C. Integer.parseInt("123")
   D. (int) "123"
9. Given String s = "Halloween";, which is true?
   A. s.length() returns 10
   B. s.substring(2,5) equals "llo"
   C. s.indexOf("allO") returns 0
   D. s.equalsIgnoreCase("HALLOWEEN") returns false
10. Which is **method overloading** (not overriding)?
    A. Same name, different parameter list, same class
    B. Same signature, subclass replaces behavior
    C. Same return type only, same class
    D. Same name in unrelated classes
11. In a well-encapsulated class, instance variables are typically…
    A. public, so clients can set them directly
    B. private, accessed via getters/setters
    C. protected, to block all outsiders including subclasses
    D. Unmodified (package-private) for performance
12. What does the this keyword help with in constructors?
    A. It imports other classes.
    B. It refers to the current object's members and can chain constructors.
    C. It makes a method static.
    D. It marks constants.
13. Which snippet **overrides** a superclass method correctly?
    A.

```
@Override
public int attackDamage() { ... }
```

    B.

```
public int attackDamage(int x) { ... }
```

C.

```
public final int attackDamage() { ... }
```

D.

```
public int attackDammage() { ... } // typo
```

14. What does super(...) do in a subclass constructor?
    A. Calls another constructor **in the same class**.
    B. Calls the superclass constructor.
    C. Calls the superclass static method.
    D. Skips field initialization.
15. Which access modifier order is from **least** to **most** visible?
    A. public → protected → default → private
    B. default → private → public → protected
    C. private → default → protected → public
    D. private → protected → default → public
16. What is printed?

```
Swordsman k = new Pillar("Kyojuro", 1000, "Fire");
System.out.println(k.receiveDamage(10));
```

Assume Pillar halves incoming damage and overrides receiveDamage.
    A. Compile error (static type mismatch)
    B. Uses Swordsman's version (full 10 damage)
    C. Uses Pillar's version (5 damage)
    D. Runtime exception

17. In Java strings, which statement is **recommended** for equality?
    A. == because it's faster
    B. .equals() for case-sensitive compare
    C. compareTo() == 0 only
    D. .hashCode() equality
18. Consider a helper to print an int[] with commas. Where should the loop live to respect method responsibilities?
    A. Inline everywhere it's needed
    B. In a dedicated printInts(int[]) static method
    C. In the constructor of the class
    D. In the array object itself
19. Which best describes a **class variable**?
    A. Declared with static, shared by all instances
    B. A final field that cannot change

C. Any private field
D. A variable declared inside a method

20. What happens if you access arr[arr.length] on an array?
    A. It returns null for the last item
    B. It wraps around to index 0
    C. Throws ArrayIndexOutOfBoundsException
    D. Returns a default value
21. Which **polymorphism** statement is true?
    A. Overloading is runtime polymorphism.
    B. Overriding enables dynamic method selection at runtime.
    C. Static methods are chosen dynamically.
    D. Fields are dynamically resolved by reference type.
22. You design receiveDamage(double dmg) to reject non-positive values. What is the **appropriate** exception to throw?
    A. ArrayIndexOutOfBoundsException
    B. ArithmeticException
    C. IllegalArgumentException
    D. NumberFormatException
23. To **catch** the exception thrown above, which header fits the catch block?
    A. catch (IllegalArgumentException e)
    B. catch (Exception())
    C. catch (IllegalArgumentException())
    D. catch (IllegalArgumentException iae) { throw; }
24. Which statement about the main method is correct?
    A. It must be an instance method.
    B. It must be public static void main(String[] args).
    C. It can return int.
    D. Parameter type can be any array of objects.
25. What does Character.isDigit(c) check?
    A. Whether c is '0' or '1' only
    B. Whether c is any numeric Unicode digit
    C. Whether c is an ASCII digit only
    D. Whether c is a letter or digit

Part 2 Code Description and Completion: 50 marks, 4 Questions.

Question 1 (10 marks)

Look at the code below, read it and then complete the code description below by filling in the spaces.

```
/** Check whether number is prime */
public static boolean isPrime() {
   Scanner sc = new Scanner(System.in);
   System.out.println("Please input a number");
   int number = 0;

   try {
      String in = sc.nextLine();
      number = Integer.parseInt(in);
   } catch (NumberFormatException e) {
      System.out.println("Non integer entered");
      System.out.println("Try again");
      return false;
   }

   for (int divisor = 2; divisor <= number / 2; divisor++) {
      if (number % divisor == 0) { // If true, number is not prime
        System.out.println("Non prime number entered");
        System.out.println("Try again");
        return false; // number is not a prime
      }
   }
   return true; // number is prime
}
```

This is a method named _____ that allows the user to test whether a number is prime. It has no parameters and returns a _____ . It is not part of an object and is a _____ method. The method first declares and initialise a _____ object, and display a message to the user to instruct them to input a number. The input will be _____in a try catch, and the String will be _____ to an integer. If the String does not parse, a _____ Exception is thrown, which will then be caught and an error message will be displayed to the user.
If the input is correctly parsed to an integer, then the method attempts to test whether the number is divisible by numbers that are less or equal than _____ of the number. It will test all candidate number with a _____ loop. Finally, if a valid input has been received, the appropriate result of true or false will be _____ from the method.

Question 2 (20 marks)

a) Create a class to store data about an Account. An Account has an id, a balance, and an annualInterestRate. You should use sensible variable types and names, and create instance variables, a constructor, appropriate encapsulation, and create a method to calculate the expected new balance after adding the interest for this year **(Note: no need to update the balance)** (10 marks)

```
public class Account {

   private int ____;
   private double balance;
   private double ____;

   public ____(int id, double balance, double annualInterestRate) {
      ____;
      this.balance = balance;
      this.annualInterestRate = annualInterestRate;
   }

      public int ____() {
          return id;
   }

      public double getBalance() {
         return ____;
      }

      public double getAnnualInterestRate() {
         return annualInterestRate;
      }

      public String toString() {
         return id + "____" + balance +"____" + annualInterestRate;
      }

      public double newBalance(){
         return ____ * (____ + annualInterestRate);
      }

}
```

b) Declare two methods named withdraw and deposit that withdraws and deposits a specified amount from the Account **(Note: the balance should be modified within the object)** (4 marks)

```
public ___ withdraw(double amount) {
   this.balance ___ amount;
}

public void ___(___) {
   this.balance += amount;
}
```

c) Create a method named transaction with the defined withdraw and deposit methods to transfer a specified amount between two accounts (the one who calls this method should be the payer, and the other is the payee), the balance of two accounts should be modified correspondingly, and the method will return true after the transaction is completed. (6 marks)

```
public ___ transaction(double ___, ___ other) {
   this.___(amount);
   other.___(amount);
   return ___;
}
```

Question 3: (10 marks)
Complete the following Java private helper method to count the occurrence of a specified character in a strings **using recursion**.

Do **not** use any loops. Submissions with loops will not pass.

Sample test cases:
String str1 = "Welcome";
System.out.println(countChar(str1,'e')); → 2
String str2 = "Good Job";
System.out.println(countChar(str2,'o')); → 3

```
// private helper method
private static int countChar(String str, char c) {
    // base case


    // recursive step


  }
```

Question 4: (10 marks)

Design a class named *Person* and its two subclasses named *Student* and *Employee* (i.e., you need to define three classes). A person must have a name, but could also has address, phone number, and e-mail address. A student has a class status (freshman, sophomore, junior, or senior). Define the status as a constant. An employee has an office and salary. You should use sensible variable types and names, and create instance variables, a constructor, appropriate encapsulation, and override the inherited methods. (10 marks)

```
class Person {
  ____ String name;

  Person(String name) {
    ____ = name;}

  public String getName() {
    return name;}
}

class Student ____ Person {
  public final static int ____ = 1;
  public final static int SOPHOMORE = 2;
  public final static int JUNIOR = 3;
  public final static int SENIOR = 4;
  protected int ____;

  Student(String name) {
    ____(name);}

  public String toString() {
    return "Student: " + ____;}
}

class ____ extends Person {
  protected String ____;
  protected int salary;

  Employee(String name) {
    super(name);}

  public ____ toString() {
    return "Employee: " + getName();}
}
```

END OF FINAL EXAM